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INSPIRING CREATIVE AND INNOVATIVE MINDS

CHAPTER 5

Finite Automata





- In computer science, we study different types of computer languages, such as Basic, Pascal, and C++.
- We will discuss a type of a language that can be recognized by special types of machines.
- Automata is a mathematical model of a machine that accepts languages of some alphabet.

- Deterministic Finite Automaton is a quintuple

$$M = \{ S, I, q_0, f_s, F \}$$

where,

S is a finite nonempty set of states

I is the input alphabet (a finite nonempty set of symbols)

q_0 is the initial state

f_s is the state transition function

F is the set of final states, subset of S .

example

- Let $M = \{ \{q_0, q_1, q_2\}, \{0, 1\}, q_0, f_s, \{q_2\} \}$

where f_s is defined as follows:

$$f_s(q_0, 0) = q_1, \quad f_s(q_1, 1) = q_2$$

$$f_s(q_0, 1) = q_0, \quad f_s(q_2, 0) = q_0$$

$$f_s(q_1, 0) = q_2, \quad f_s(q_2, 1) = q_1$$

- Note that for M :
 $S = \{q_0, q_1, q_2\}$, $I = \{0, 1\}$, $F = \{q_2\}$
 q_0 is the initial state

S is a finite nonempty set of states
 I is the input alphabet (a finite nonempty set of symbols)
 q_0 is the initial state
 f_s is the state transition function
 F is the set of final states, subset of S .

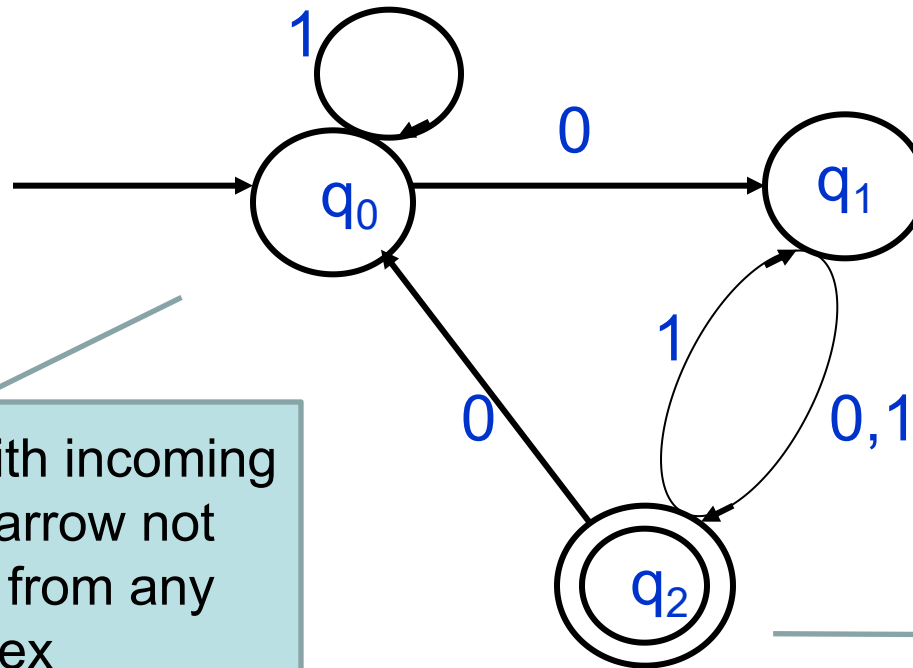
example

- The state transition function of a DFA is often described by means of a table, called a **transition table**.

f_s	0	1
q_0	q_1	q_0
q_1	q_2	q_2
q_2	q_0	q_1

example

- The **transition diagram** of this DFA is,



Each state represented by a small circle labeled with the state

Initial state with incoming unlabeled arrow not originating from any vertex

Final state with a double circle

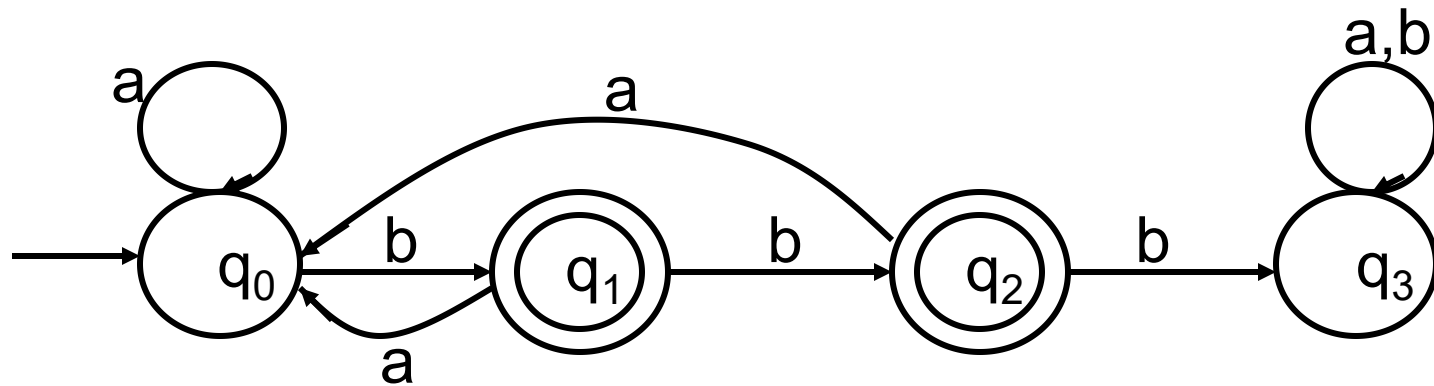
Let $M = (\{q_0, q_1, q_2, q_3\}, \{a, b\}, q_0, f_s, \{q_1, q_2\})$
where f_s is given by the table

f_s	a	b
q_0	q_0	q_1
q_1	q_0	q_2
q_2	q_0	q_3
q_3	q_3	q_3

- S is a finite nonempty set of states
- I is the input alphabet (a finite nonempty set of symbols)
- q_0 is the initial state
- f_s is the state transition function
- F is the set of final states, subset of S .

example

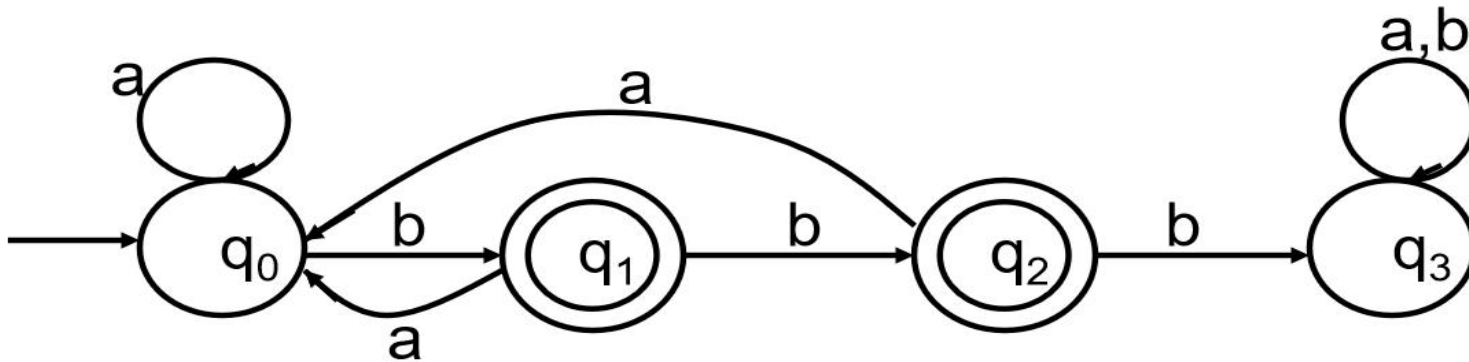
- The transition diagram of this DFA is,





- Let $M = \{ S, I, q_0, f_s, F \}$ be a DFA and w is an input string,
- w is said to be accepted by M if $f_s^*(q_0, w) \in F$
- f_s^* - extended transition function for M

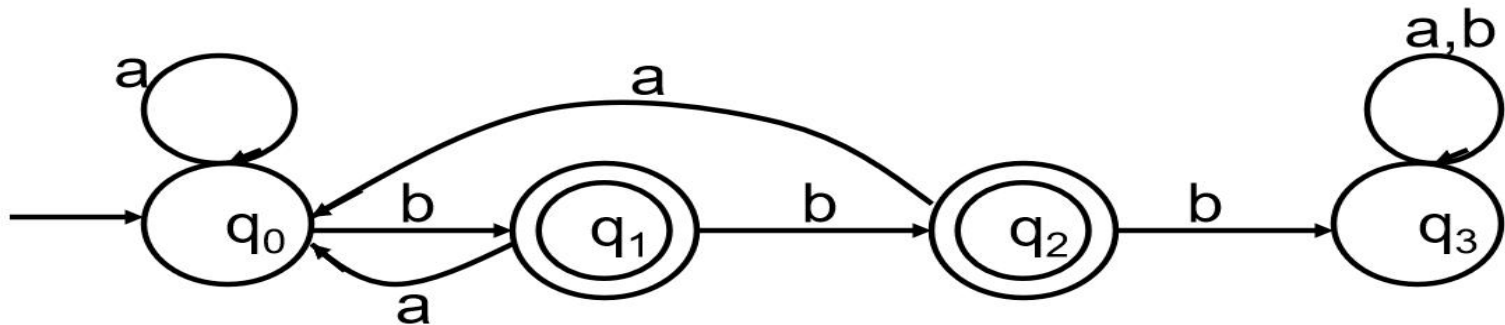
$w = abb$



$q_0 \xrightarrow{a} q_0 \xrightarrow{b} q_1 \xrightarrow{b} q_2$

accepted
by M

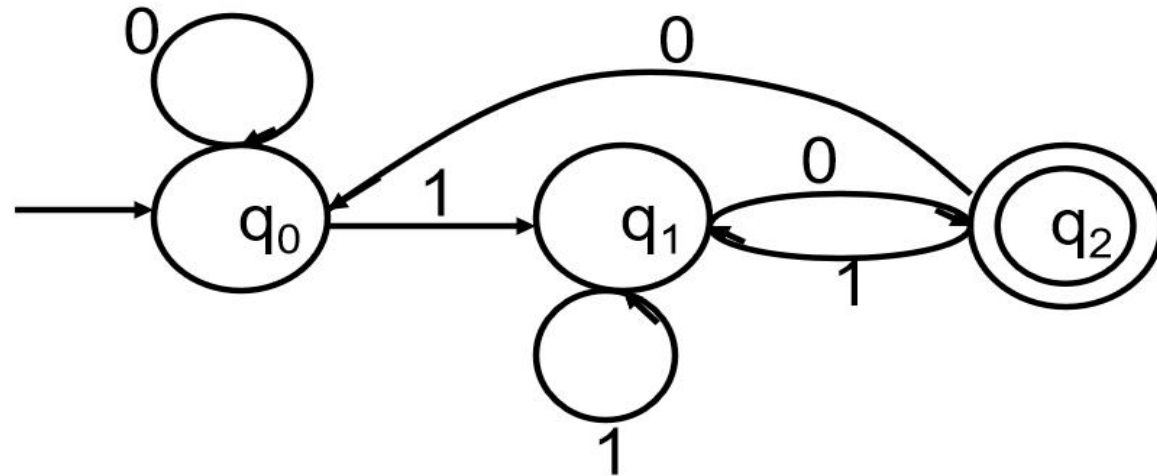
$w = abba$



$q_0 \xrightarrow{a} q_0 \xrightarrow{b} q_1 \xrightarrow{b} q_2 \xrightarrow{a} q_0$

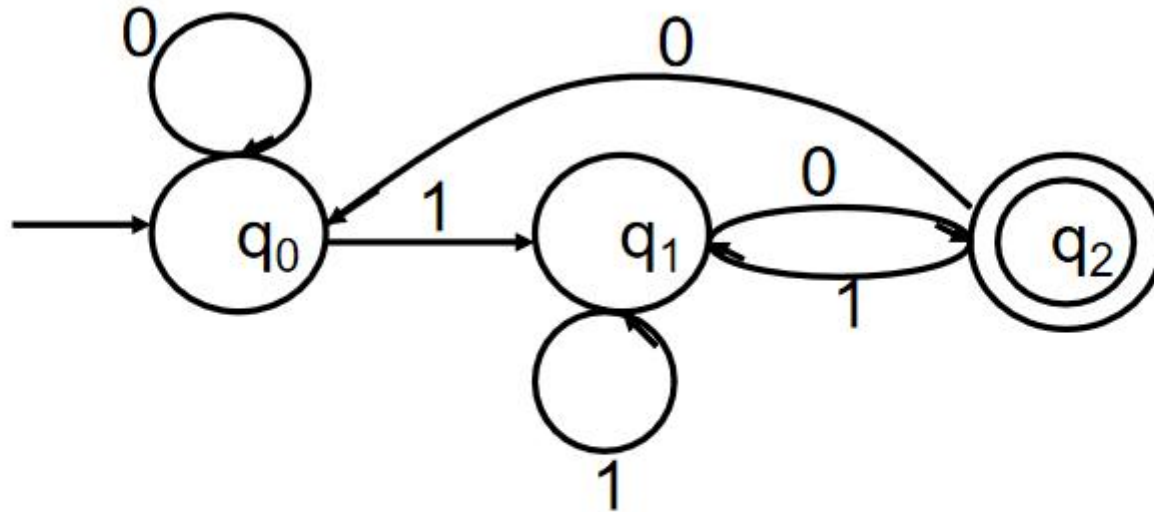
not accepted by M

example



- What are the states of M? q_0, q_1, q_2
- Write the set of input symbols. $I = \{0, 1\}$
- Which is the initial state? q_0

example

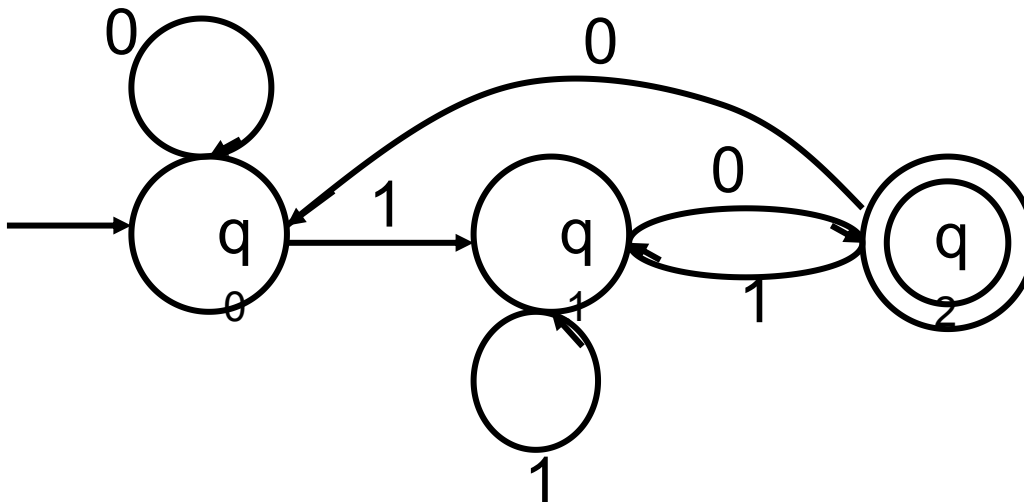


● Write the set of final states.

$F = \{q_2\}$

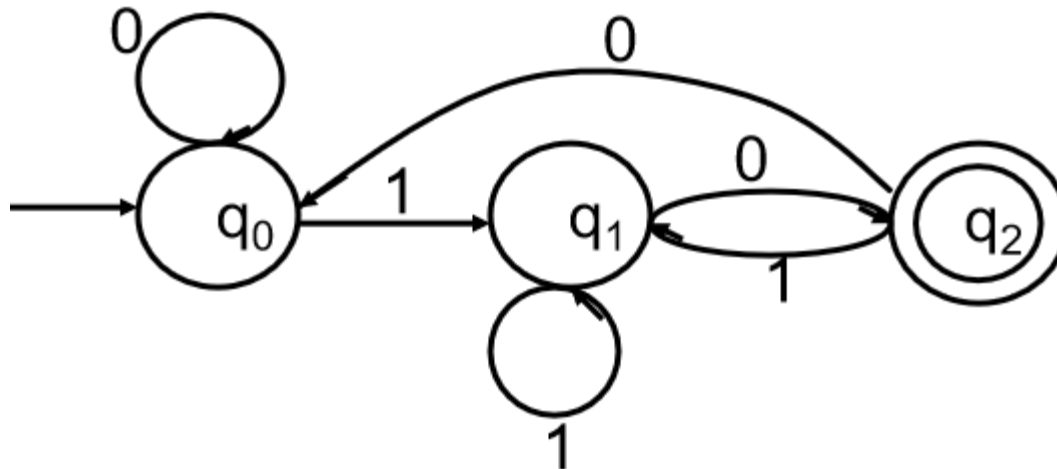
example

Write the transition table for this DFA



f_s	0	1
q_0	q_0	q_1
q_1	q_2	q_1
q_2	q_0	q_1

example



Which of the strings are accepted by M?

0111010, 00111, 111010, 0100, 1110

✓

✗

✓

✗

✓

example

Construct a state transition diagram of a DFA that accepts on $\{a,b\}$ that contain an even number of a's and an odd number of b's.

Example of accepted strings:

aab, baa, baaabba

F	a	b
q_0		
q_1		
q_2		
q_3		

states and input

q_0	even num. of a's & even num. of b's.
q_1	even num. of a's & odd num. of b's.
q_2	odd num. of a's & odd num. of b's.
q_3	odd num. of a's & even num. of b's.

set of states, $S = \{q_0, q_1, q_2, q_3\}$

initial state, q_0

final state, q_1

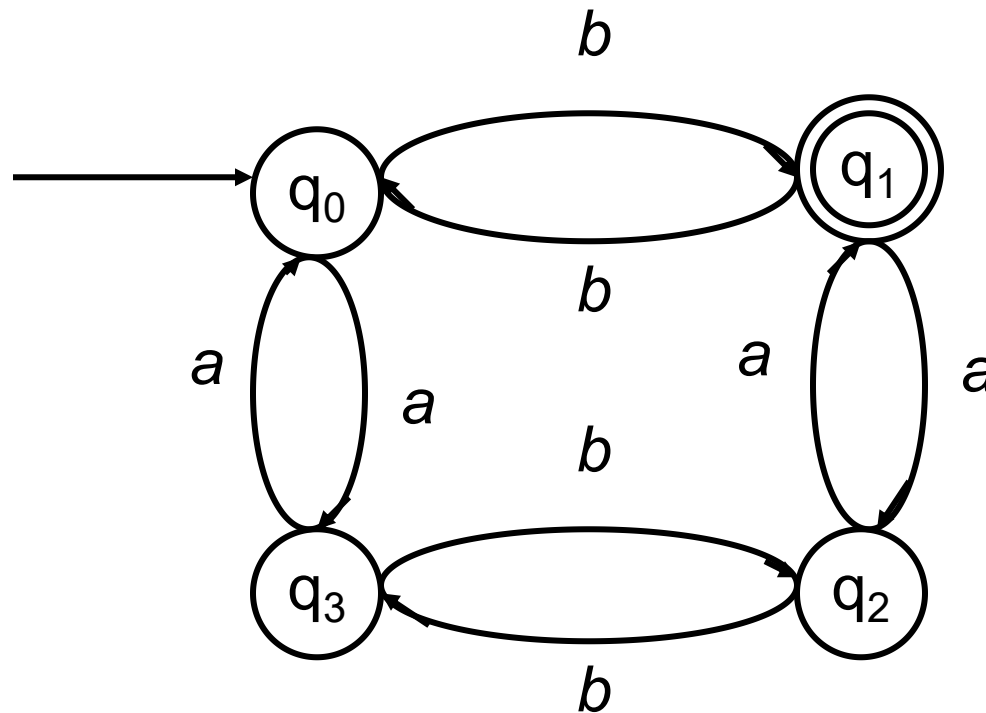
set of input symbols, $I = \{a, b\}$



state transition function

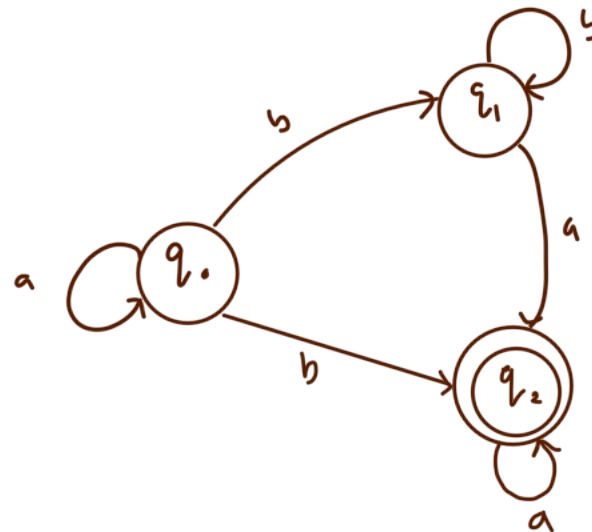
f_s	a	b
q_0	q_3	q_1
q_1	q_2	q_0
q_2	q_1	q_3
q_3	q_0	q_2

State transition diagram



Let $M=(S, I, q_0, f_s, F)$ be the DFA such that $S=\{q_0, q_1, q_2\}$, $I=\{a, b\}$, $F=\{q_2\}$, q_0 =initial state, and f_s is given by transition table below.

f_s	a	b
q_0	q_0	q_1
q_1	q_2	q_1
q_2	q_2	q_0



- Draw the state diagram of M.
- Which of the strings **abaa**, **bbbabb**, **bbbbaa** dan **bababa** are accepted by M?

- Automata with input as well as output.
- Every state has an input and corresponding to the input the state also has an output.
- These types of automata are commonly called **finite state machines**.
- A finite state machine is a sextuple,

$M = \{ S, I, O, q_0, f_s, f_o \}$, where

S	is a finite nonempty set of states
I	is the input alphabet
O	is the output alphabet
q_0	is the initial state
f_s	is the state transition function
f_o	is the output function.



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- Let $M = \{ S, I, O, q_0, f_s, f_o \}$ be the FSM
- where,
 - $S = \{q_0, q_1, q_2\},$
 - $I = \{a, b\},$
 - $O = \{0, 1\},$
 - $q_0 =$ initial state,



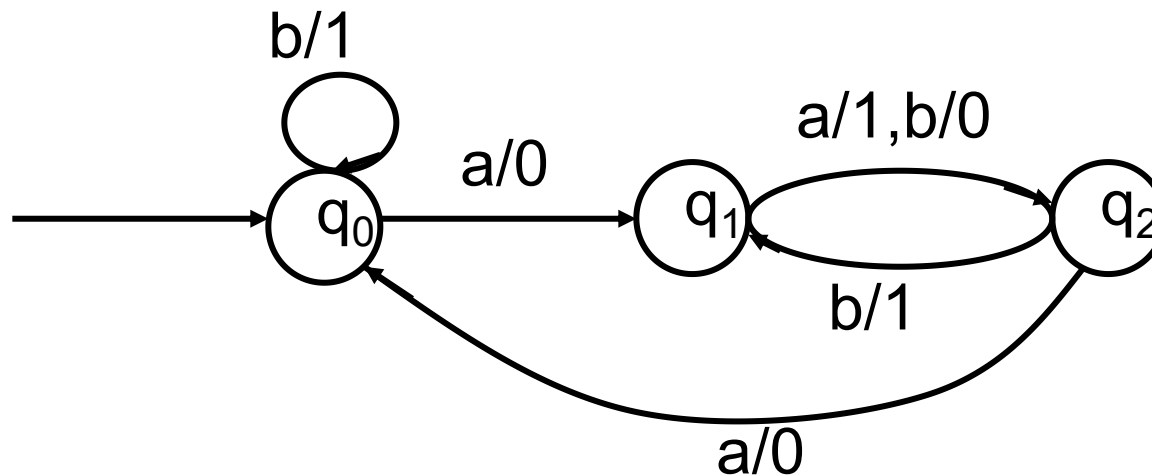
FSM state transition table

f_s and f_o

	f_s		f_o	
	a	b	a	b
q_0	q_1	q_0	0	1
q_1	q_2	q_2	1	0
q_2	q_0	q_1	0	1

example

Given the transition digaram of FSM below.
Find the output string and output when input
string **bbab** and **bababaa**

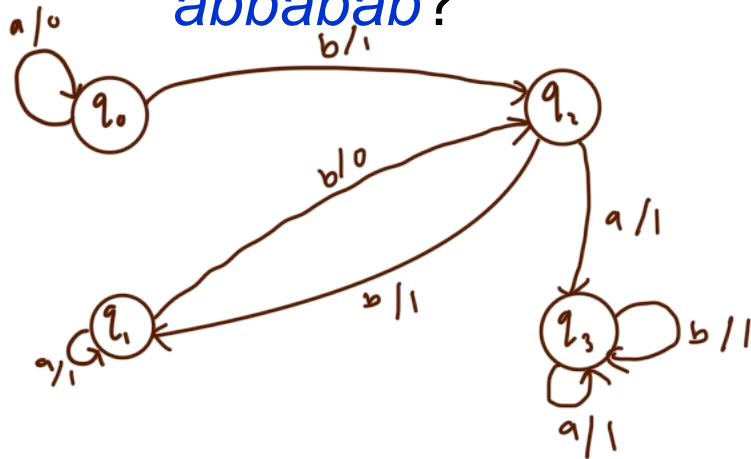




- Let $M = \{ S, I, O, q_0, f_s, f_o \}$ be the FSM
- where,
 $S = \{q_0, q_1, q_2, q_3\}$,
 $I = \{a, b\}$,
 $O = \{0, 1\}$,
 q_0 = initial state,

example

- Given the f_s and f_o
- Draw the transition diagram of M.
- What is the **output string** and the **output** if the input string is *abbabab*?



	f_s		f_o	
	a	b	a	b
q_0	q_0	q_2	0	1
q_1	q_1	q_2	1	0
q_2	q_3	q_1	1	1
q_3	q_3	q_3	1	1

OS: 0111011

O: 1

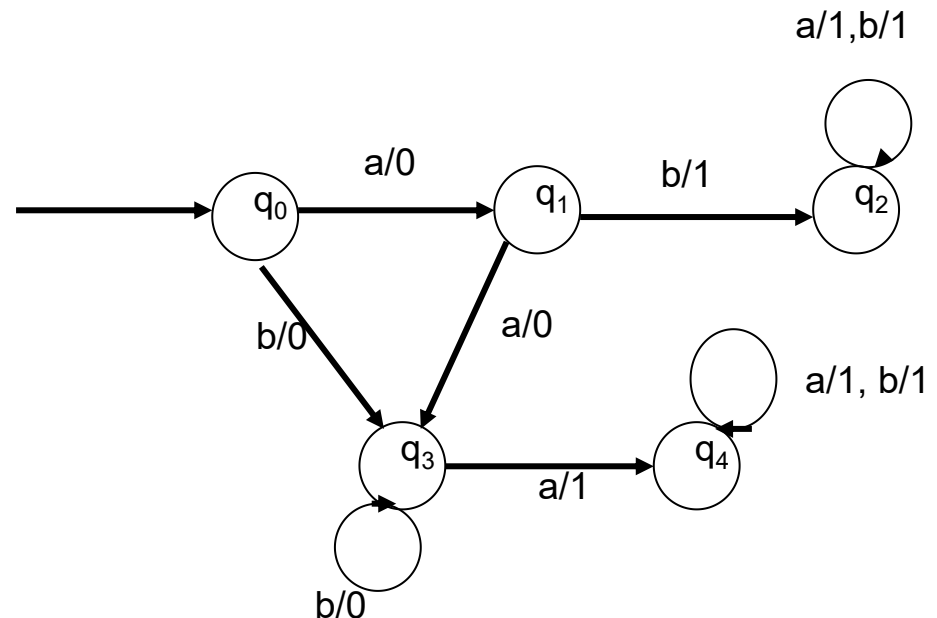
$q_0 \rightarrow q_0 \rightarrow q_2 \rightarrow q_1 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3 \rightarrow q_3$



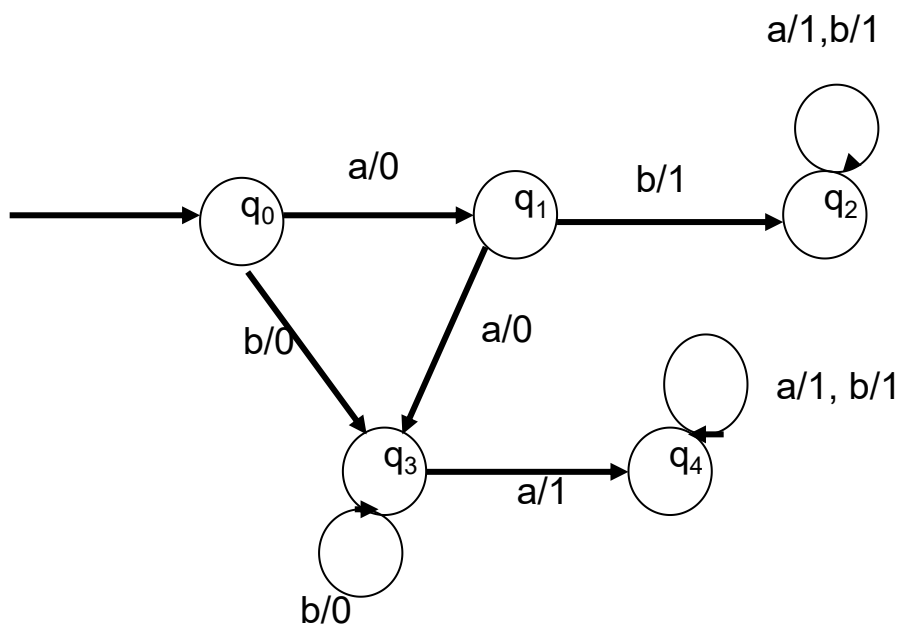
- Let M be a FSM.
- Let x be a nonempty string in M .
- We say that x is accepted by M if and only if the output of x is 1.

example

- Write the transition table of M.
- What is the output string if the input string is *aaabbbb*? *001111*
- What is the output if the input string is *bbbaaaa*? *1*
- Is the string *aaa* accepted by M? ✓



The transition table of M.



	f_s		f_o	
	a	b	a	b
q_0	q_1	q_3	0	0
q_1	q_3	q_2	0	1
q_2	q_2	q_2	1	1
q_3	q_4	q_3	1	0
q_4	q_4	q_4	1	1

exercise

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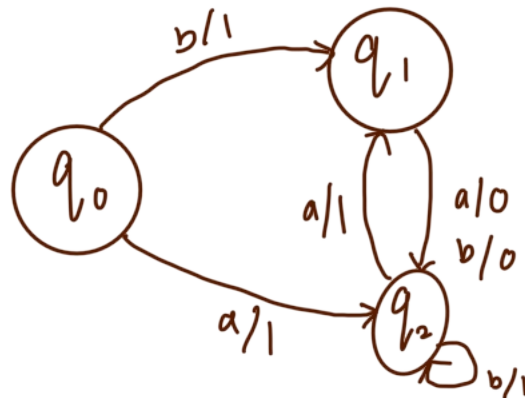
Let $M = \{ S, I, O, q_0, f_s, f_o \}$ be a FSM
where,

$S = \{q_0, q_1, q_2\}$,

$I = \{a, b\}$,

$O = \{0, 1\}$,

q_0 = initial state,



	f_s		f_o	
	a	b	a	b
q_0	q_2	q_1	1	1
q_1	q_2	q_2	0	0
q_2	q_1	q_2	1	1

- Draw the transition diagram of M.
- What is the output string if the input string is aabbb? = 11011
- What is the output string if the input string is ababab? = 111010
- What is the output if the input string is abbbaba? = 1
- What is the output if the input string is bbbababa? = 1

prepared by Razana Alwee



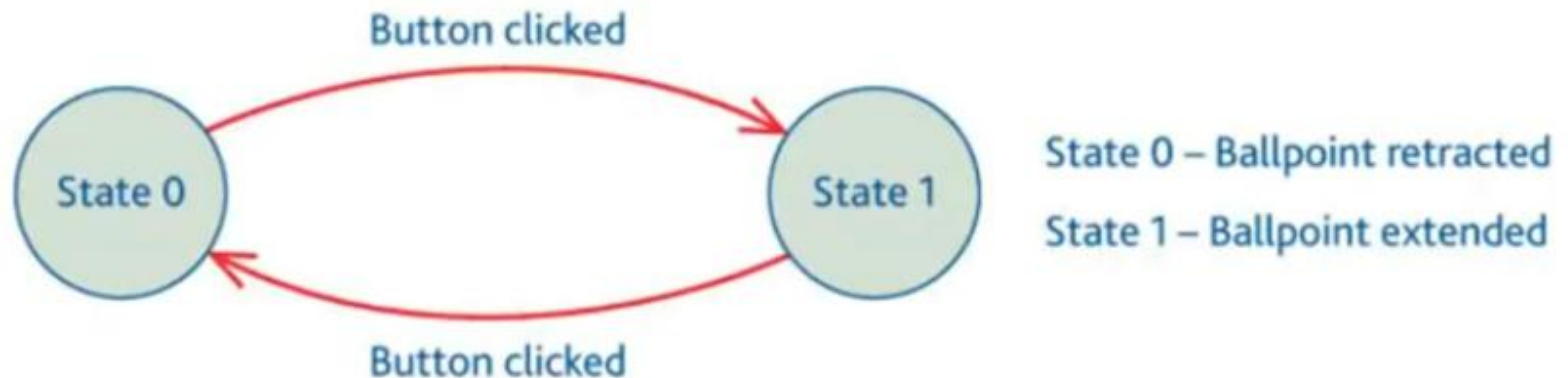
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APPLICATIONS OF FINITE AUTOMATA

Ballpoint pen

- Finite States:
 - Ballpoint retracted, Ballpoint extended
- Inputs: Click the button



Combination Lock

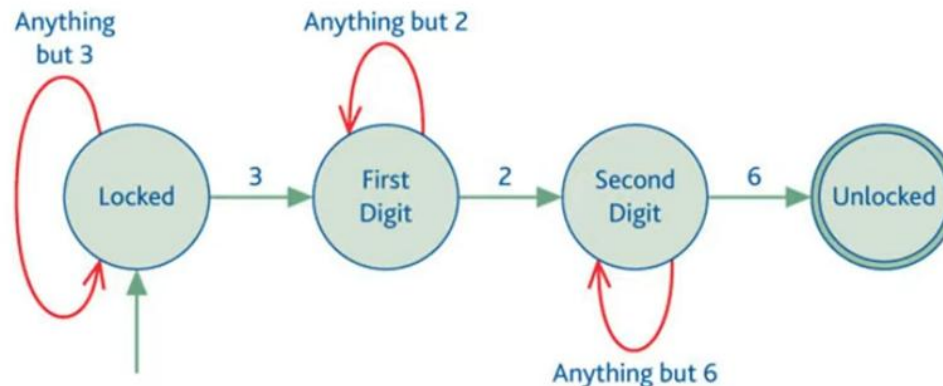
- Unlocked when the correct input, 326, is entered.

State transition table



Input	Current State	Next state
Anything but 3	Locked	Locked
3	Locked	First Digit
Anything but 2	First Digit	First Digit
2	First Digit	Second Digit
Anything but 6	Second Digit	Second Digit
6	Second Digit	Unlocked

State transition diagram



Vending Machine

- The vending machine will dispense a candy priced at 1.25.

